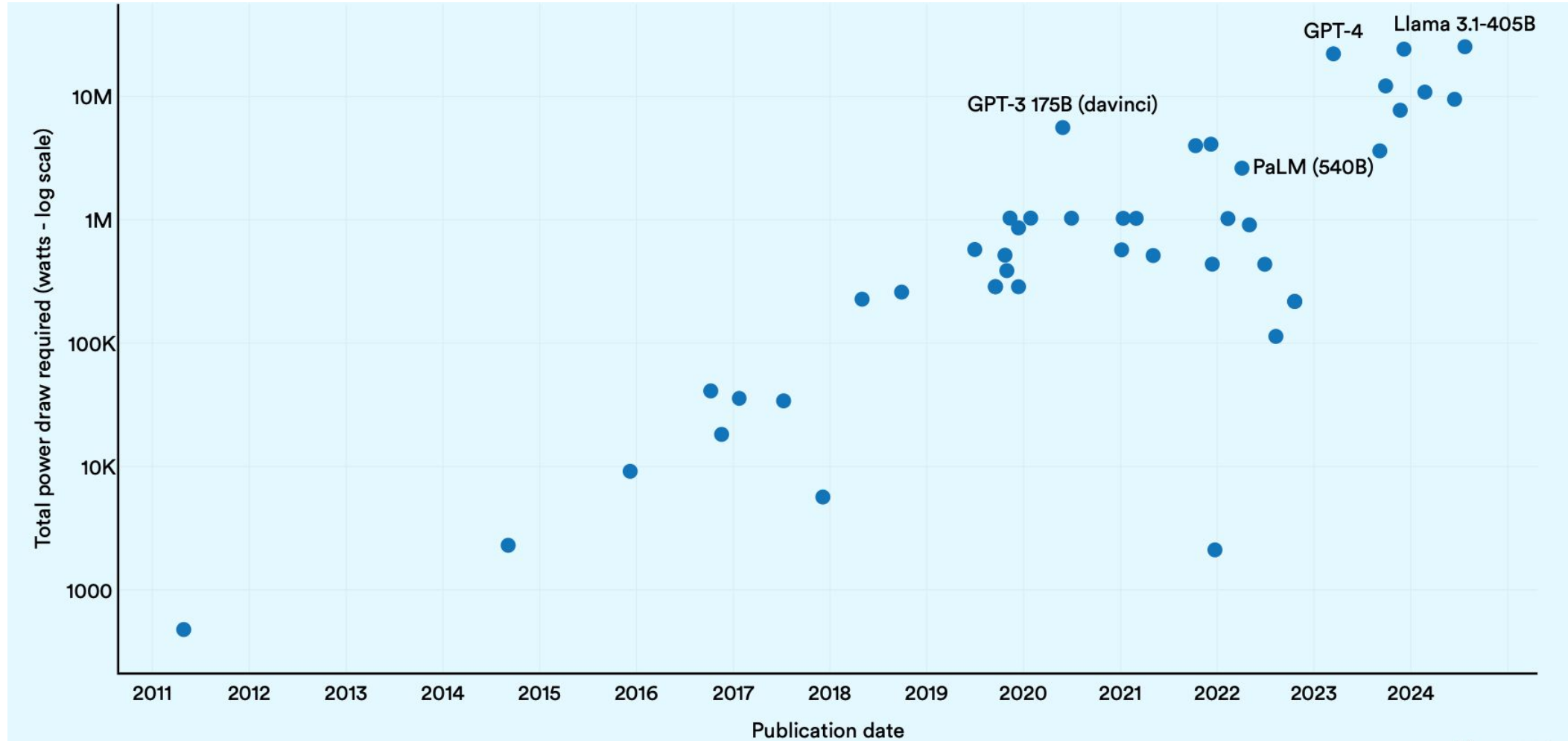


Efficient AI

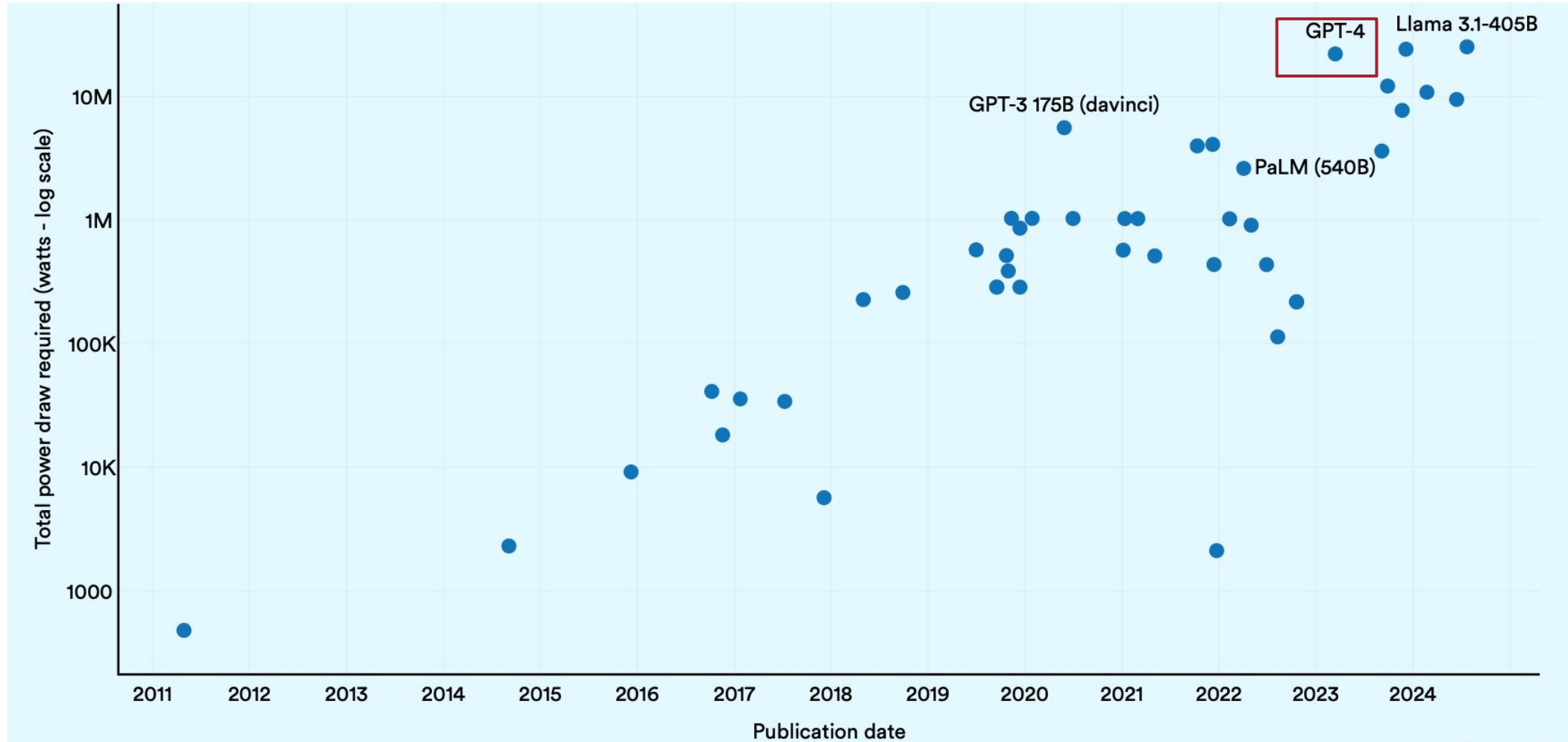
Guosheng Hu

Power Consumption of Model Training

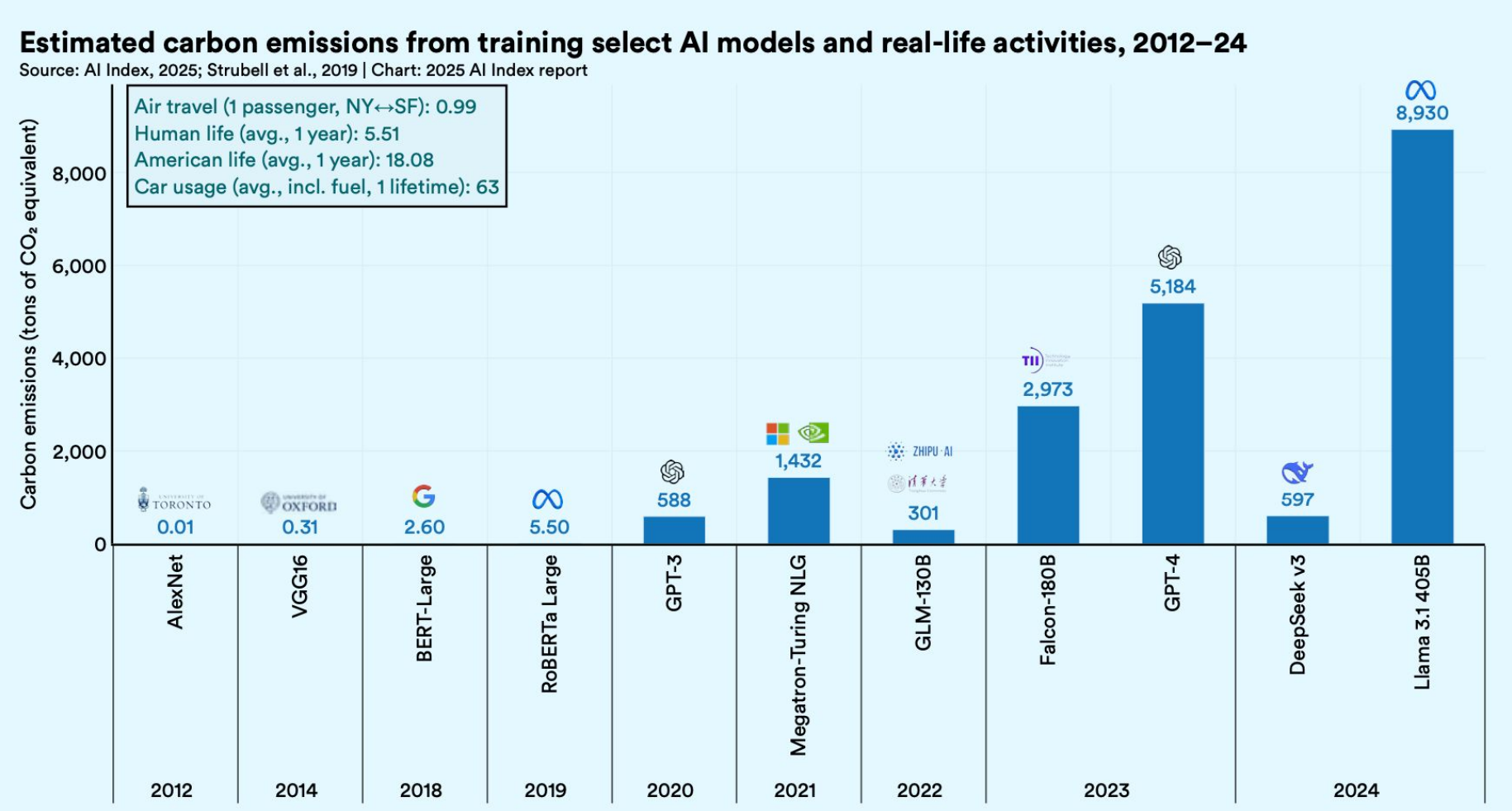


Power Consumption of Model Training

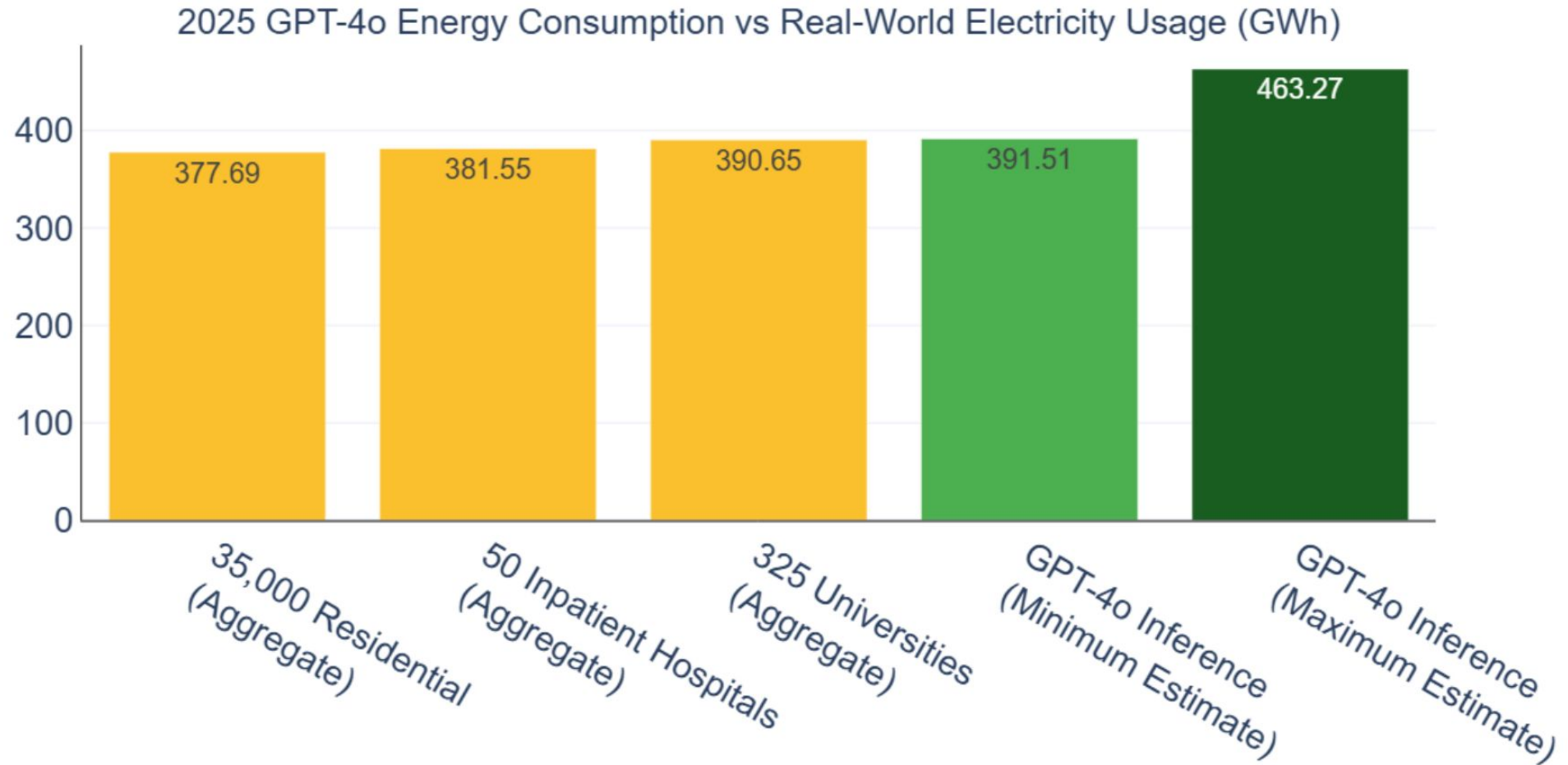
> \$100M



Carbon Emission of Model Training

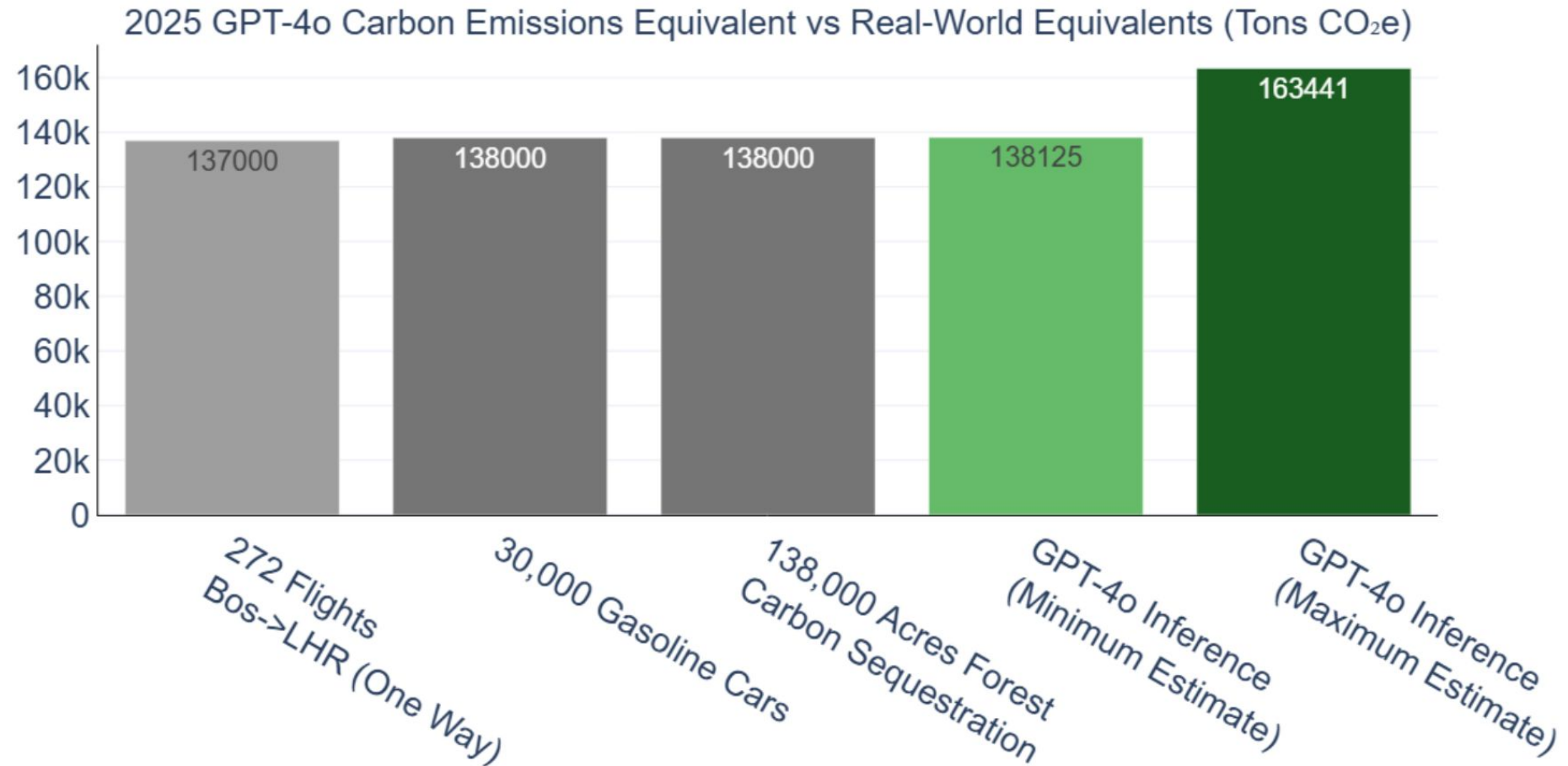


Power Consumption of Model Inference



Jegham, Nidhal, et al. "How hungry is ai? benchmarking energy, water, and carbon footprint of llm inference." *arXiv preprint arXiv:2505.09598* (2025).

Carbon Emission of Model Inference



Jegham, Nidhal, et al. "How hungry is ai? benchmarking energy, water, and carbon footprint of llm inference." *arXiv preprint arXiv:2505.09598* (2025).

Efficient AI Solutions

- Model Quantisation
- Model Pruning
- Architecture Design (e.g. Mamba)
- Hardware

Gu, Albert, and Tri Dao. "Mamba: Linear-time sequence modeling with selective state spaces." *First conference on language modeling*. 2024.

Efficient AI Solutions

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Quantisation - Numeric Data Types

Integer

- Unsigned Integer
 - n -bit Range: $[0, 2^n - 1]$
- Signed Integer
 - Sign-Magnitude Representation
 - n -bit Range: $[-2^{n-1} + 1, 2^{n-1} - 1]$
 - Both 000...00 and 100...00 represent 0
 - Two's Complement Representation
 - n -bit Range: $[-2^{n-1}, 2^{n-1} - 1]$
 - 000...00 represents 0
 - 100...00 represents -2^{n-1}

0	0	1	1	0	0	0	1
x	x	x	x	x	x	x	x

$$2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0 = 49$$

Sign Bit

1	0	1	1	0	0	0	1
x	x	x	x	x	x	x	x

$$- 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0 = -49$$

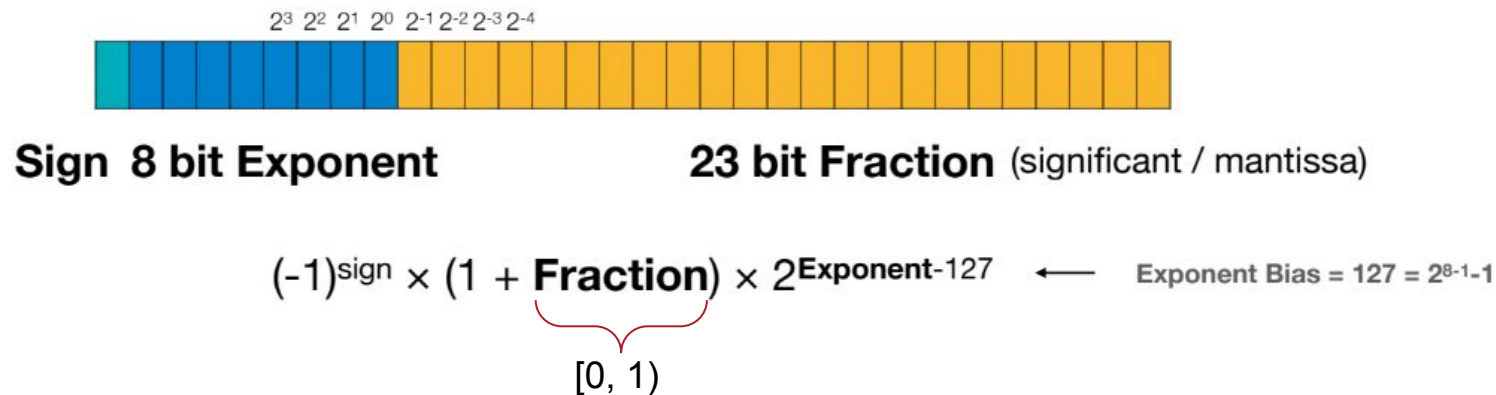
1	1	0	0	1	1	1	1
x	x	x	x	x	x	x	x

$$-2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0 = -49$$

Quantisation - Numeric Data Types

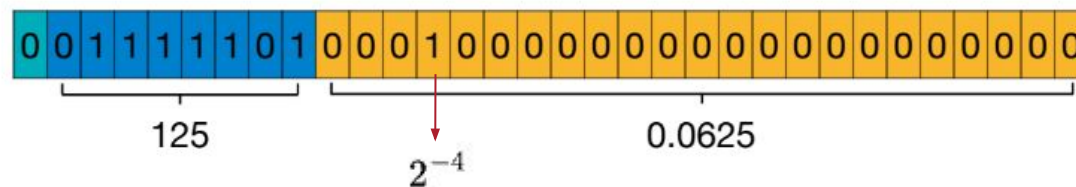
Floating-Point Number

Example: 32-bit floating-point number in IEEE 754



How to represent **0.265625**?

$$0.265625 = 1.0625 \times 2^{-2} = (1 + \underline{0.0625}) \times 2^{125-127}$$



Quantisation - Numeric Data Types

Floating-Point Number

Exponent Width → Range; Fraction Width → Precision

[IEEE 754](#) Single Precision 32-bit Float (IEEE FP32)



Exponent
(bits)

8

Fraction
(bits)

23

Total
(bits)

32

[IEEE 754](#) Half Precision 16-bit Float (IEEE FP16)



5

10

16

[Google](#) Brain Float (BF16)



8

7

16

[Nvidia](#) FP8 (E4M3)



* FP8 E4M3 does not have INF, and $S.1111.111_2$ is used for NaN.
* Largest FP8 E4M3 normal value is $S.1111.110_2 = 448$.

4

3

8

[Nvidia](#) FP8 (E5M2) for gradient in the backward



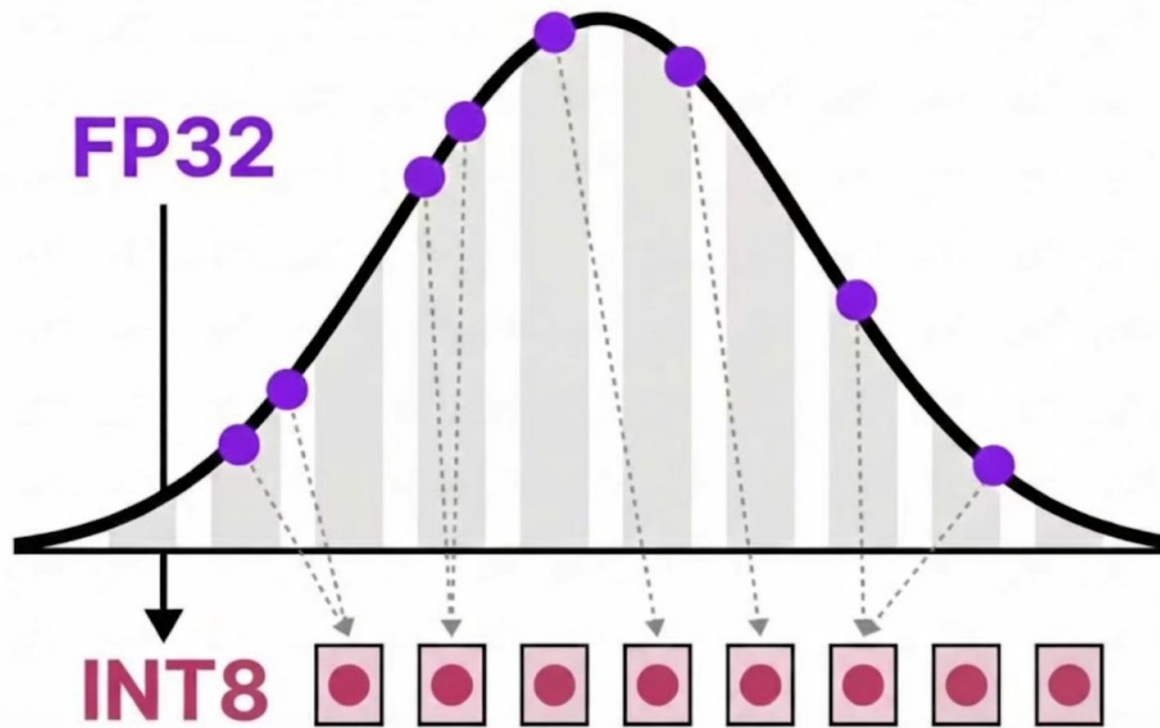
* FP8 E5M2 have INF ($S.11111.00_2$) and NaN ($S.11111.XX_2$).
* Largest FP8 E5M2 normal value is $S.11110.11_2 = 57344$.

5

2

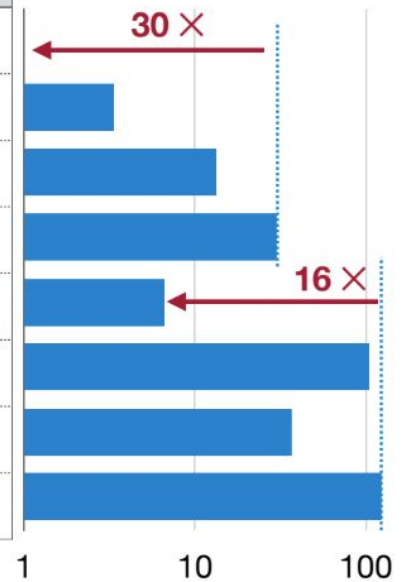
8

Motivation



Operation	Energy [pJ]
8 bit int ADD	0.03
32 bit int ADD	0.1
16 bit float ADD	0.4
32 bit float ADD	0.9
8 bit int MULT	0.2
32 bit int MULT	3.1
16 bit float MULT	1.1
32 bit float MULT	3.7

Rough Energy Cost For Various Operations in 45nm 0.9V



Quantisation - Linear Quantisation

Quantisation equation: $q = \text{Round}\left(\frac{r}{S}\right) + Z$

-4	80	127
16	55	-34
-128	5	23

Quantised matrix (q)
(INT8)

=

Round(

0.34	3.75	5.64
1.12	2.7	-0.9
-4.7	0.68	1.43

Real matrix (r)
(FP32)

/ 0.04055) + (-12)

Scale (S)
(FP32)

Quantisation parameter

Zero-point (Z)
(INT8)

Quantisation parameter

Linear Quantisation - Scale

Quantisation equation: $q = \text{Round}\left(\frac{r}{S}\right) + Z$

$$S = \frac{r_{\max} - r_{\min}}{q_{\max} - q_{\min}}$$

- $r_{\max}$ and $r_{\min}$: max/min of the real matrix
- $q_{\max}$ and $q_{\min}$: max/min of the quantised data type

0.34	3.75	5.64
1.12	2.7	-0.9
-4.7	0.68	1.43

Real matrix
(FP32)

Quantising to INT8 :

$$r_{\max} = 5.64, \quad r_{\min} = -4.7$$

$$q_{\max} = 2^7 - 1 = 127, \quad q_{\min} = -2^7 = -128$$

$$S = \frac{r_{\max} - r_{\min}}{q_{\max} - q_{\min}} = \frac{5.64 - (-4.7)}{127 - (-128)} = 0.04055$$

Linear Quantisation - Zero Point

Quantisation equation: $q = \text{Round}\left(\frac{r}{S}\right) + Z$

$$Z = q_{\min} - \text{Round}\left(\frac{r_{\min}}{S}\right)$$

- $r_{\min}$: min of the real matrix
- $q_{\min}$: min of the quantised data type

0.34	3.75	5.64
1.12	2.7	-0.9
-4.7	0.68	1.43

Real matrix
(FP32)

Quantising to INT8 :

$$r_{\max} = 5.64, \quad r_{\min} = -4.7$$

$$q_{\max} = 2^7 - 1 = 127, \quad q_{\min} = -2^7 = -128$$

$$S = \frac{r_{\max} - r_{\min}}{q_{\max} - q_{\min}} = \frac{5.64 - (-4.7)}{127 - (-128)} = 0.04055$$

$$Z = q_{\min} - \text{Round}\left(\frac{r_{\min}}{S}\right) = -128 - \text{Round}\left(-\frac{4.7}{0.04055}\right) = -12$$

Linear Quantisation - De-Quantisation

De-Quantisation equation: $r' = S(q - Z)$

$$\begin{pmatrix} -4 & 80 & 127 \\ 16 & 55 & -34 \\ -128 & 5 & 23 \end{pmatrix} - (-12) \times 0.04055 = \begin{pmatrix} 0.32 & 3.73 & 5.64 \\ 1.14 & 2.72 & -0.89 \\ -4.7 & 0.69 & 1.42 \end{pmatrix}$$

Quantised matrix (q)
(INT8)

Zero-point (Z)
(INT8)
Quantisation parameter

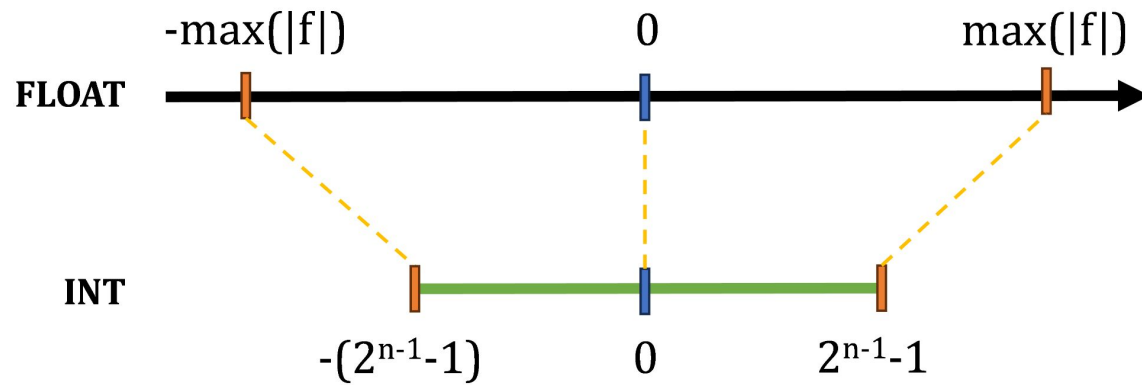
Scale (S)
(FP32)
Quantisation parameter

Reconstructed matrix (r')
(FP32)

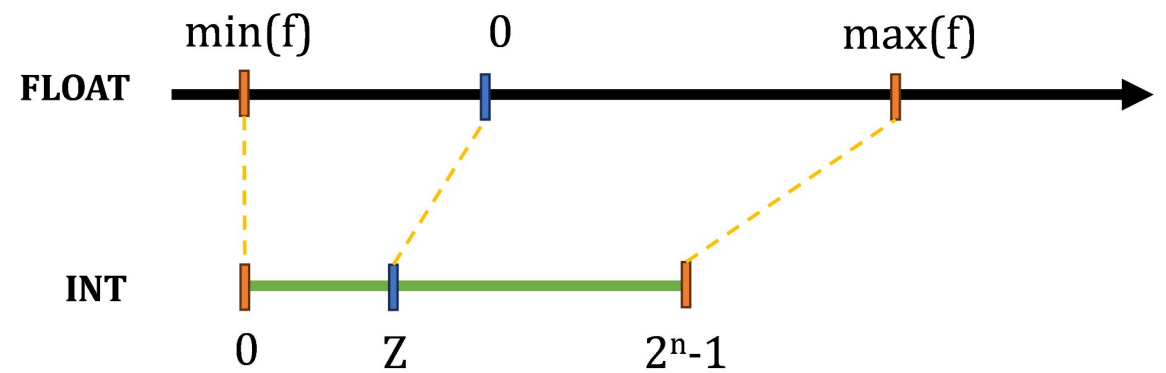
Quantisation error = Real matrix — Reconstructed matrix =

0.02	0.02	0
-0.02	-0.02	-0.01
0	-0.01	0.01

Symmetric vs Asymmetric Linear Quantisation



Symmetric quantisation



Asymmetric quantisation

Symmetric Linear Quantisation

Quantisation equation: $q = \text{Round}\left(\frac{r}{S}\right) + Z$, where $Z=0$

7	84	127
25	61	-20
-106	15	32

= Round(

0.34	3.75	5.64
1.12	2.7	-0.9
-4.7	0.68	1.43

/ 0.04441)

Quantised matrix (q)
(INT8)

Real matrix (r)
(FP32)

Scale (S)
(FP32)

Quantisation parameter

Scale parameter: $S = \frac{|r|_{max}}{q_{max}} = \frac{5.64}{127} = 0.04441$

De-Quantisation equation: $r' = S \times q$

Weight (W) and Activation (A)

For neural network quantisation, both weights and activations can be quantised.

Consider the following matrix multiplication:

$$Y = WX$$

After weight & activation quantisation:

Output in original
data space

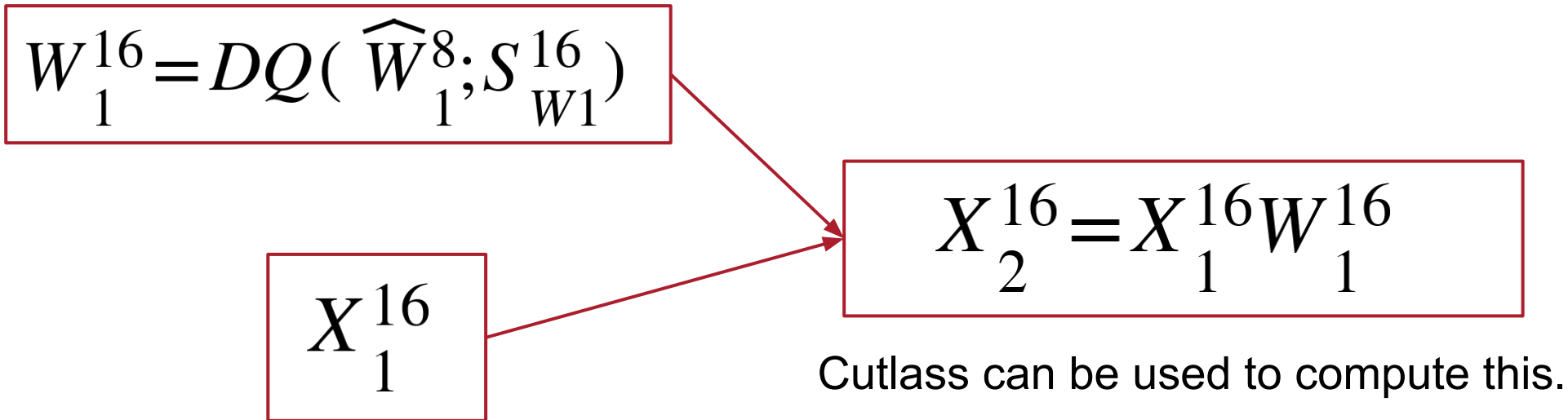
$$\begin{aligned} Y' &= Dequant(q_w) \times Dequant(q_x) \\ &= S_w(q_w - Z_w) \times S_x(q_x - Z_x) \\ &= S_w S_x(q_w q_x - Z_w q_x - Z_x q_w + Z_w Z_x) \end{aligned}$$

Inference of Quantised Neural Networks

1. The scale is calculated and saved. For example, the scale for weight or activation quantisation FP16 to INT8 are S_X^{16} and S_W^{16} , which are both FP16.
2. Before a neural network inference, the quantised weights are pre-calculated $\hat{W}^8 = Q(W^{16}; S_W^{16})$;
3. During the inference, the quantised weights $\hat{W}^8$ are loaded and used for multiplications with Activation X .
4. W8A16 and W8A8 are widely used.

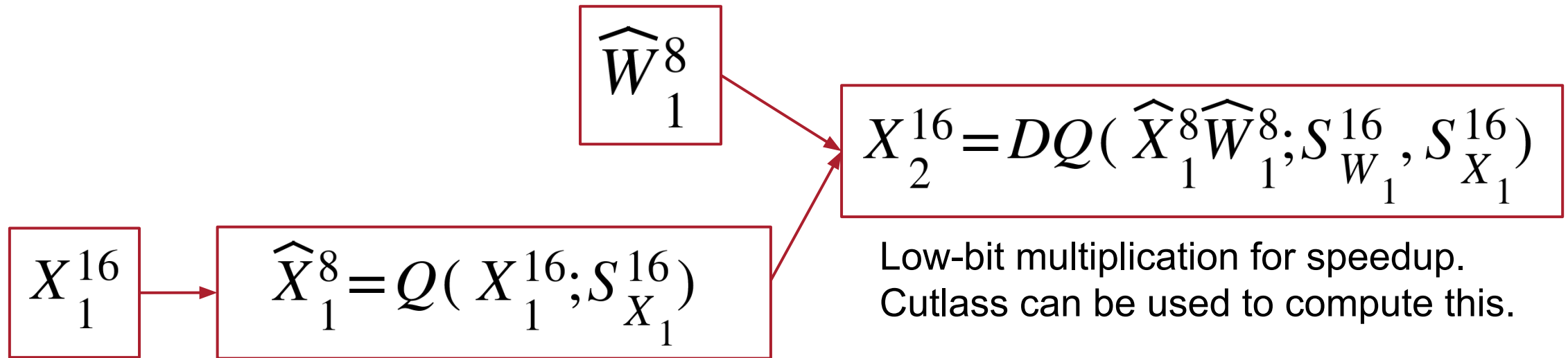
One Implementation of W8A16 Inference

DeQuant W to FP16 to do Matrix Multiplication in FP16.



One Implementation of W8A8 Inference

Quantise X and do Matrix Multiplication in INT8, and then Dequant:



Low-bit multiplication for speedup.
Cutlass can be used to compute this.

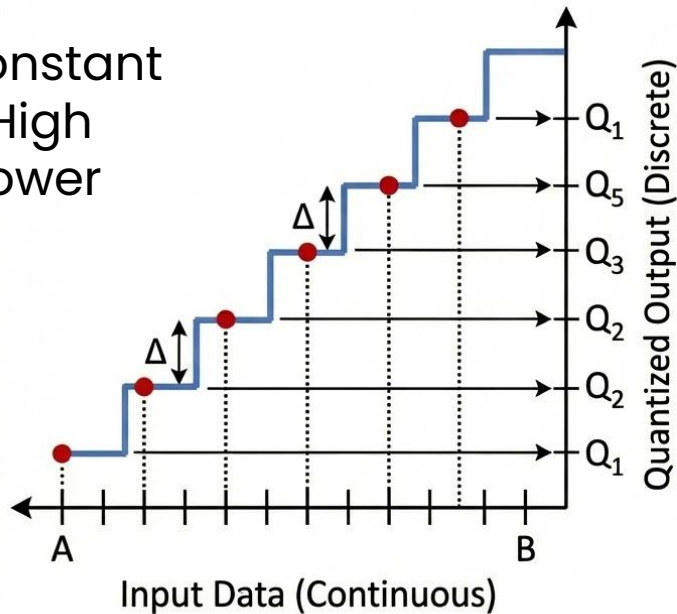
Uniform Vs. Non-Uniform Quantisation

Uniform Quantisation

discrete values with equal spacing

Uniform Quantization

step size: constant
Simplicity: High
Precision: Lower

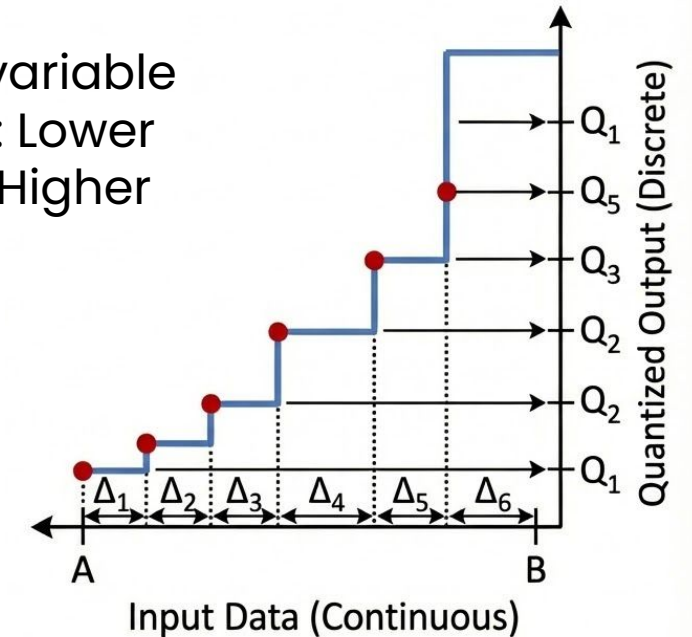


Non-uniform Quantisation

discrete values with unequal spacing

Non-Uniform Quantization

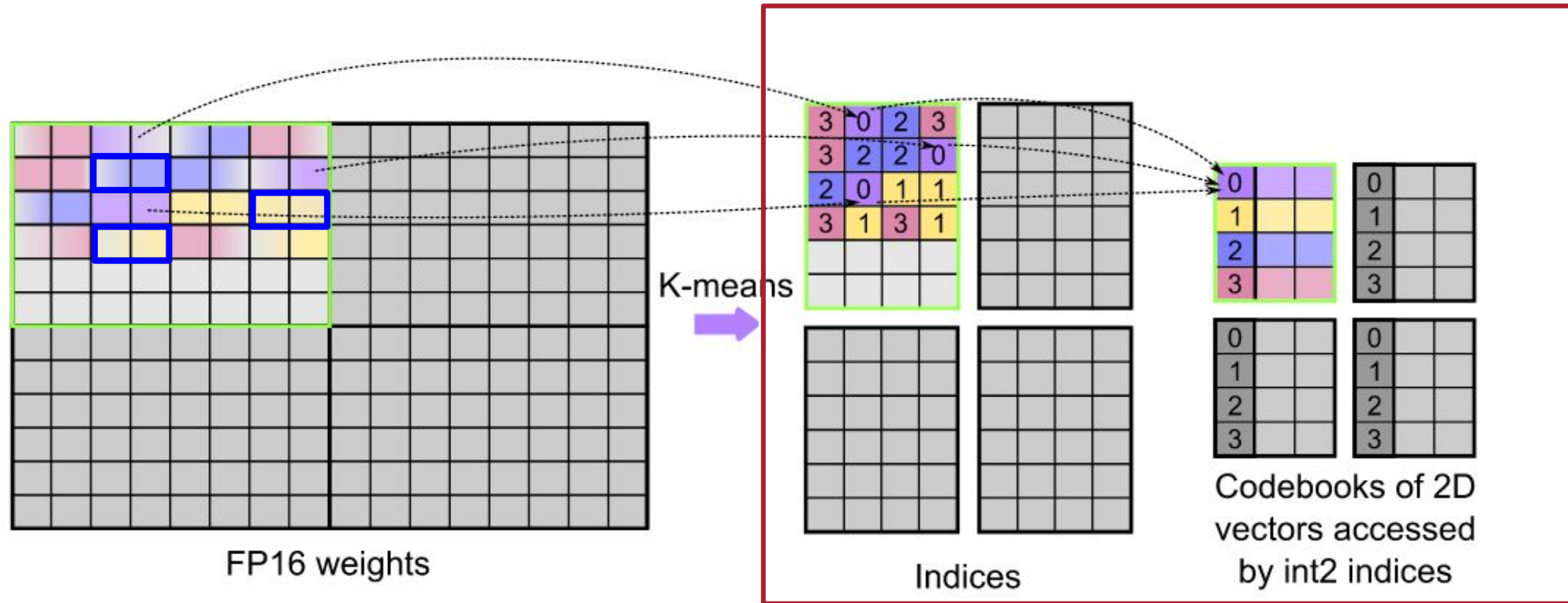
step size: variable
Simplicity: Lower
Precision: Higher



Vector Quantisation (Non-Uniform Quantisation)

Offline VQ Process (2D vector, K=2)

stored in memory

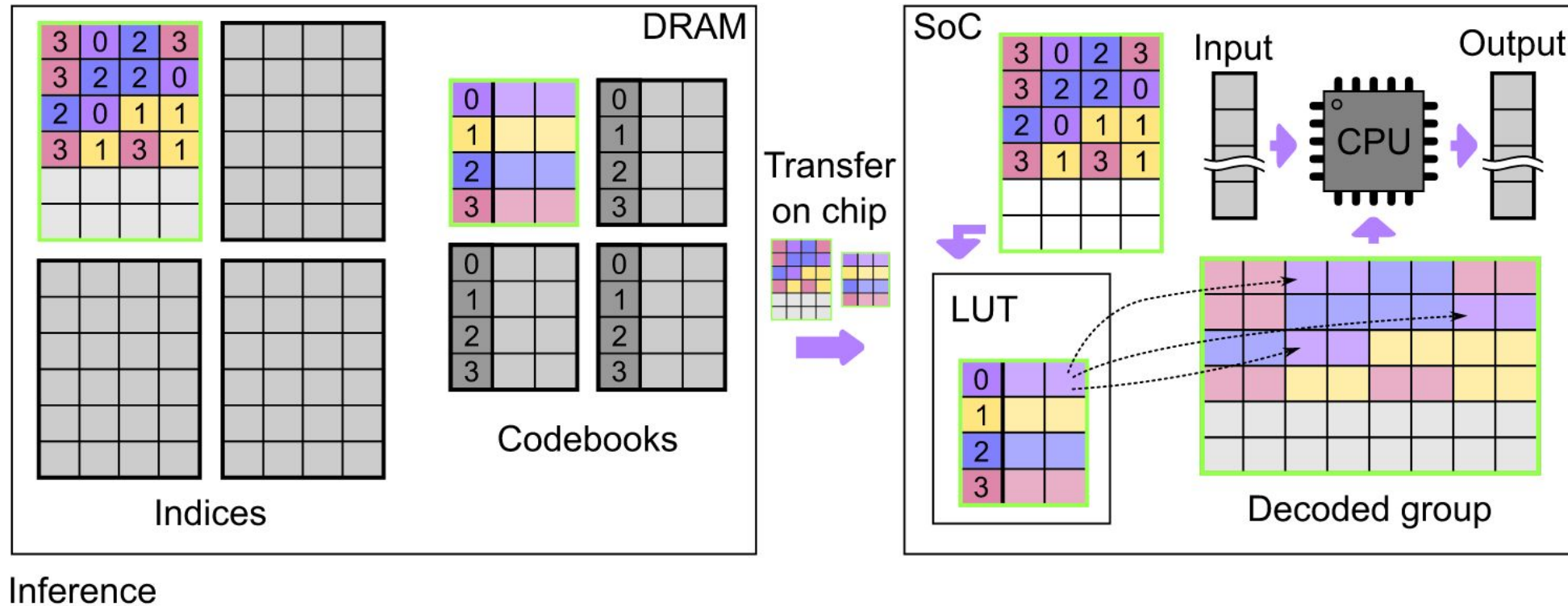


Index 3 represent the values in 3rd centroid of K-means. Then 6 x 8 FP16 matrix become 6 x 4 Int 2 matrix. Int 2 is enough for 4 clusters. codebook stores the centroid.

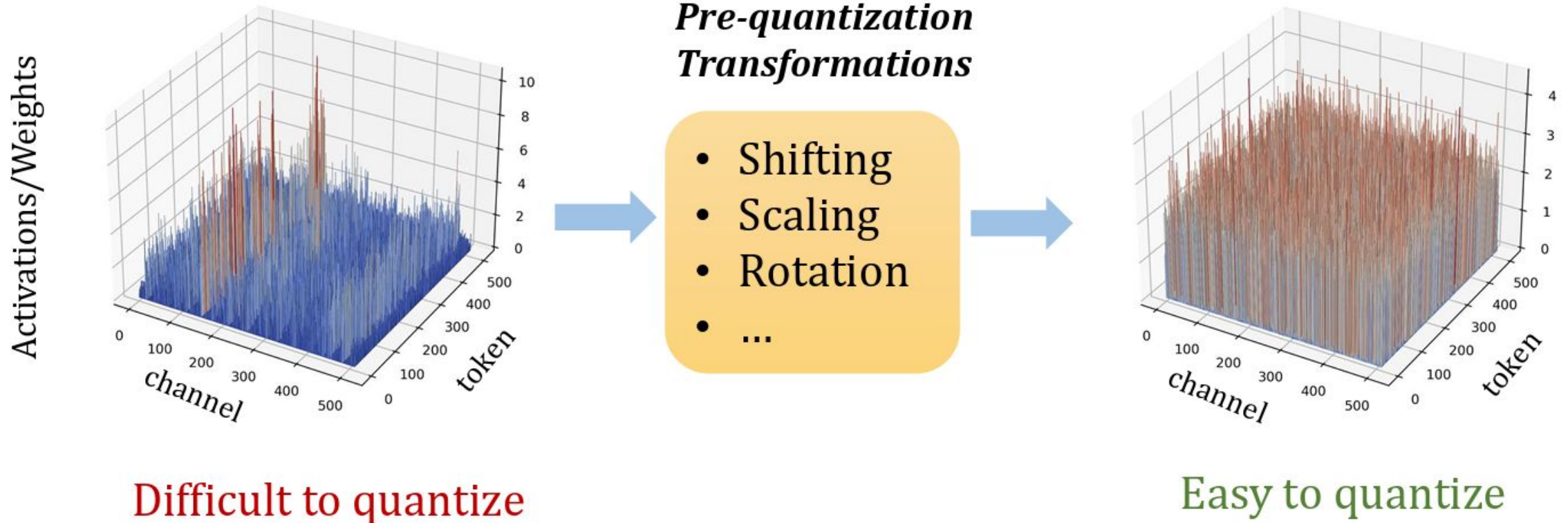
Vector Quantisation

DeQuant Process

Online inference Process



Challenges of LLM Quantisation



Liu, Yutong, Cairong Zhao, and Guosheng Hu. "A Comprehensive Evaluation on Quantization Techniques for Large Language Models." *arXiv preprint arXiv:2507.17417* (2025).